Dynamics around Charged Black Holes in Entangled Relativity

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Plan

Introduction

The two principles of General Relativity:

- General covariance
- Equivalence principle

Conventions: $(-,+,+,+)$ metric signature and $c = \hbar = G = 4\pi\varepsilon_0 = 1$ natural system of units.

Einstein-Hilbert action:

$$S\left[g,\Psi\right]=\int\left(\frac{R}{2\kappa}+\mathcal{L}_m\right)\sqrt{|g|}\mathrm{d}^4x \tag{1}$$

Einstein field equations:

$$R_{\mu\nu}-\frac{1}{2}Rg_{\mu\nu}=\kappa T_{\mu\nu} \tag{2}$$

with $T_{\mu\nu}=-\frac{2}{\sqrt{|g|}}\frac{\partial(\sqrt{|g|}\mathcal{L}_m)}{\partial g^{\mu\nu}}$.

Reissner-Nordström Black Hole

Reissner-Nordström solution:

$$\begin{aligned} \mathrm{d}s^2 = &- \left(1 - \frac{r_+}{r}\right)\left(1 - \frac{r_-}{r}\right)\mathrm{d}t^2 \\ &+ \left(1 - \frac{r_+}{r}\right)^{-1}\left(1 - \frac{r_-}{r}\right)^{-1}\mathrm{d}r^2 \\ &+ r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\mathrm{d}\varphi^2\right) \end{aligned} \tag{3}$$

Horizons:

$$r_\pm = M \pm \sqrt{M^2 - Q^2} \tag{4}$$

The solution exists for $|Q|/M \leq 1$.

Entangled Relativity

Action of Entangled Relativity:

$$S\left[g,\Psi\right]=\int\frac{\mathcal{L}_m^2}{R}\sqrt{|g|}\mathrm{d}^4x \tag{5}$$

Field equations:

$$R_{\mu\nu}-\frac{1}{2}Rg_{\mu\nu}=-\frac{R}{\mathcal{L}_m}T_{\mu\nu}+\frac{R^2}{\mathcal{L}_m^2}\left(\nabla_\mu\nabla_\nu-g_{\mu\nu}\Box\right)\frac{\mathcal{L}_m^2}{R^2} \tag{6}$$

Scalar-Tensor form of Entangled Relativity

Charged Black Holes in Entangled Relativity

Charged black hole solution of ER:

$$\begin{aligned} \mathrm{d}s^2 = &- \left(1 - \frac{r_+}{r}\right)\left(1 - \frac{r_-}{r}\right)^{15/13} \mathrm{d}t^2 \\ &+ \left(1 - \frac{r_+}{r}\right)^{-1}\left(1 - \frac{r_-}{r}\right)^{-7/13} \mathrm{d}r^2 \\ &+ r^2 \left(1 - \frac{r_-}{r}\right)^{6/13} \left(\mathrm{d}\theta^2 + \sin^2\theta \mathrm{d}\varphi^2\right) \end{aligned} \tag{7}$$

Scalar Field:

$$\phi = \left(1 - \frac{r_-}{r}\right)^{-4/13} \tag{8}$$

Conformal Transformation

Einstein frame obtained by:

$$g_{\mu\nu} \mapsto \tilde{g}_{\mu\nu} = \phi g_{\mu\nu} \tag{9}$$

New solution:

$$\begin{aligned}
\mathrm{d}\tilde{s}^2 = &- \left(1 - \frac{r_+}{r}\right)\left(1 - \frac{r_-}{r}\right)^{11/13} \mathrm{d}t^2 \\
&+ \left(1 - \frac{r_+}{r}\right)^{-1}\left(1 - \frac{r_-}{r}\right)^{-11/13} \mathrm{d}r^2 \\
&+ r^2 \left(1 - \frac{r_-}{r}\right)^{2/13} \left(\mathrm{d}\theta^2 + \sin^2\theta \mathrm{d}\varphi^2\right)
\end{aligned} \tag{10}$$

Domain of Validity of the Solution

Horizons:

$$r_- = \frac{13}{11}\left(M - \sqrt{M^2 - \frac{11}{12}Q^2}\right) \tag{11}$$

$$r_+ = M + \sqrt{M^2 - \frac{11}{12}Q^2} \tag{12}$$

Two conditions:

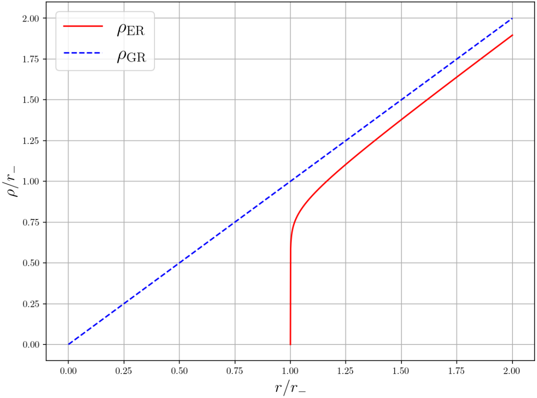
- Real-valued radii: $|Q|/M \leq \sqrt{12/11}$
- Inner horizon lower than outer horizon: $r_- \leq r_+$

Solution exists for $|Q|/M \leq \sqrt{143/132}$.

Areal Radius

Areal radius:

$$\rho_{\mathrm{GR}} = r_{\mathrm{GR}}, \quad \rho_{\mathrm{ER}} = r_{\mathrm{ER}} \left(1 - \frac{r_-}{r_{\mathrm{ER}}}\right)^{1/13} \tag{13}$$



Null Geodesics Analysis

Null Geodesics Analysis

Timelike Geodesics Analysis

Timelike Geodesics Analysis

Black Hole Phenomenology

Conclusion